Synthetic Instance Configuration 200–600

Generated Files

Instance file	$\|S\|$	$\|P\|$	$\|B\|$	$\|E\|$	$\|T\|$	$\|I\|$	Seed
problem_instance_ext_200_0.txt	200	240	33	4	12	33	1234
problem_instance_ext_200_1.txt	200	240	26	6	14	26	1235
problem_instance_ext_200_2.txt	200	240	33	5	6	33	1236
problem_instance_ext_200_3.txt	200	240	26	2	2	26	1237
problem_instance_ext_400_0.txt	400	480	69	5	13	26	1238
problem_instance_ext_400_1.txt	400	480	57	3	7	57	1239
problem_instance_ext_400_2.txt	400	480	57	2	3	57	1240
problem_instance_ext_400_3.txt	400	480	53	3	3	53	1241
problem_instance_ext_600_0.txt	600	720	90	2	2	90	1242
problem_instance_ext_600_1.txt	600	720	85	5	6	85	1243
problem_instance_ext_600_2.txt	600	720	79	4	8	79	1244
problem_instance_ext_600_3.txt	600	720	77	3	6	77	1245

$|S|$: students; $|P|$: placements; $|B|$: companies; $|E|$: educational centres; $|T|$: professors; $|I|$: dependency groups; Seed: seed used to generate the instance.

Generator Arguments

Argument	Description	Value
result_path	Output folder where the generated files are written	output
start_size	First number of students considered	200
end_size	Last number of students considered	600
number_instances	Number of instances generated for each student size	4
surplus	Extra placements percentage with respect to students	20
slack_ratio	Extra capacity margin for company and professor quotas	0.2
placement_dep_prob	Percentage used to create placement dependency groups	30
number_competences	Number of base competences per profile	6
most_important_competences	Number of primary competences required by a placement	2
seed	Base seed used to derive the seed of each generated instance	1234

Internal Parameters

Parameter	Description	Value
student_size_step	Fixed step between generated student sizes	200
number_profiles	Number of academic profile blocks	8
p_excellent_student	Probability of generating an excellent student	0.05
p_notable_student	Probability of generating a notable student	0.25
car_probability	Probability that a student has access to a car	0.65
student_distance_weight_range	Range used for the student distance weight	$[0.0, 0.5]$
n_centers_min	Minimum number of educational centres	2
n_centers_max	Maximum number of educational centres	6
prof_capacity_min	Minimum professor supervision capacity	5
prof_capacity_max	Maximum professor supervision capacity	25
ORS_AVAILABLE	Whether ORS routing support is available in the script	False
use_ors	Whether ORS is actually used for this generation	False
ors_base_url	ORS endpoint used when routing is enabled	http://localhost:8080/ors
synthetic_duration_mu	Mean of the synthetic duration distribution	1500
synthetic_duration_sigma	Standard deviation of the synthetic duration distribution	250

Randomness

Field	Description	Value
Base seed	Initial seed provided to the generator	1234
Seed policy	Rule used to derive instance seeds from the base seed	instance_seed = base_seed + global_instance_idx
Python random seeded	Whether the Python random generator is seeded per instance	Yes
NumPy random seeded	Whether the NumPy random generator is seeded per instance	Yes